

Math 1220 Summer Exam 1 Review

1. Let $f(x) = x^2 - 3x - 4$ and $h(x) = \frac{2}{x+1}$. Find the following.

Evaluate. (a) $f(x + 3)$

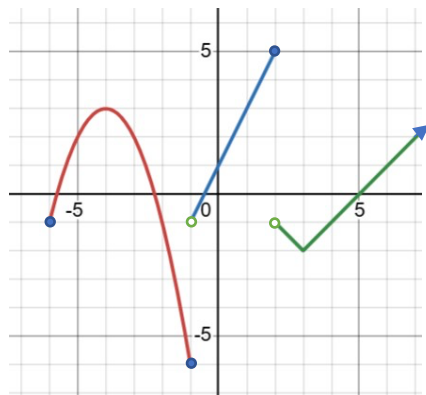
(b) $h\left(\frac{4}{w}\right)$

Solve. (c) $f(x) = 0$

(d) $h(x) = 3$

(e) Find the intercepts of each function and write them as ordered pairs.

2. Use the graph of $y = f(x)$:



(a) $f(0) =$

$f(2) =$

$f(5) =$

(b) Find all x for which: $f(x) = 3$

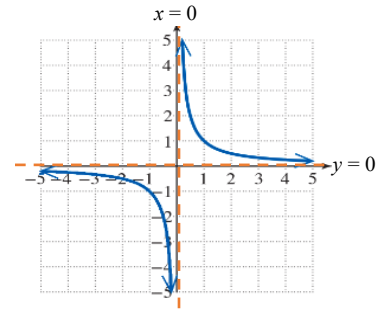
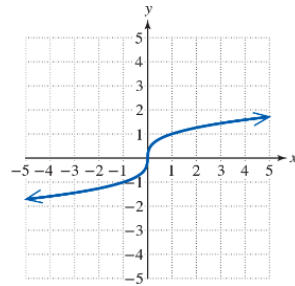
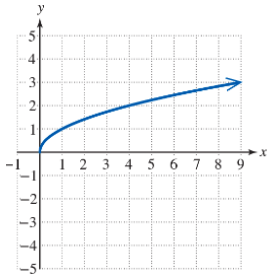
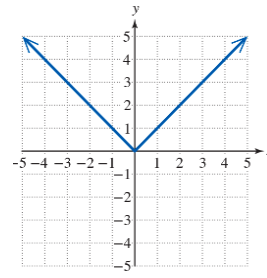
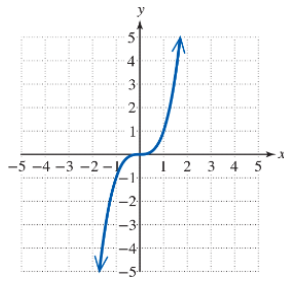
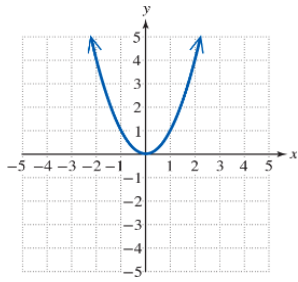
$f(x) = -1$

(d) Domain of f _____

Range of f _____

e) Find the interval where function is increasing _____

3. Write the equation of each function.



4. Determine the transformation being applied to the basic curve.

(a) $y = 3|x|$

(b) $y = -\sqrt{x}$

(c) $y = \frac{1}{x} - 3$

(d) $y = (x + 3)^2$

(e) $y = \sqrt[3]{-x}$

5. Determine all the intercepts and transformations being applied to the basic curve.

(f) $f(x) = 1 - \frac{1}{2}(x + 4)^2$

(g) $g(x) = \sqrt{2 - x} - 3$

6. Determine the domain for each function. Write your answers using interval notation.

(a) $f(x) = (x + 3)(x - 1)$

(b) $g(x) = \frac{x+2}{(x+3)(x-1)}$

(c) $h(x) = \sqrt{2x - 3}$

(d) $j(x) = \frac{\sqrt{2x-3}}{x+2}$

(d) $g(x) = \frac{x+2}{\sqrt{3-2x}}$

(e) $g(x) = \frac{x-6}{\sqrt{3-2x}-4}$

(f) $g(x) = \frac{6}{|3-2x|-4}$

7. Find the slope of the line that is perpendicular to the line $2x - 3y = 1$.

8. Find the equation of the line that passes through the points (1, -2) and (3, 4) Write your answer in slope-intercept form.

9. Find the equation of the line that passes through the points (1, -2) and parallel to $2x - 3y = 6$. Write your answer in slope-intercept form.

10. Let $f(x) = 2x^2 - 3x + 1$.

(a) Find the average rate of change of over the interval $[-1, 3]$.

(b) Evaluate and simplify $\frac{f(x+h)-f(x)}{h}$

11. Let $g(x) = \frac{3}{x}$. Evaluate and simplify $\frac{g(1+h)-g(1)}{h}$.

12. Graph $f(x) = 3 - 2(x - 1)^3$ by applying transformations in the indicated stages. Indicate 3 points on each curve.

Basic curve

Stretch/Shrink

Reflection

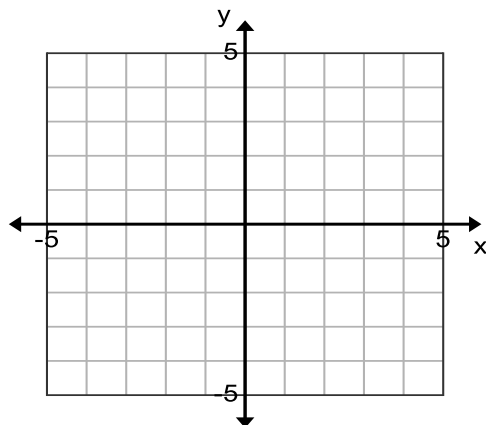
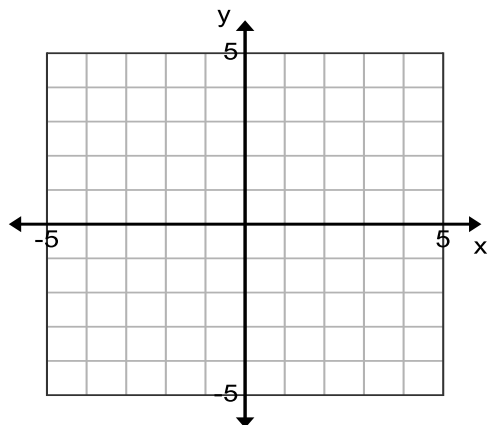
Shifts: _____

$y = \underline{\hspace{2cm}}$

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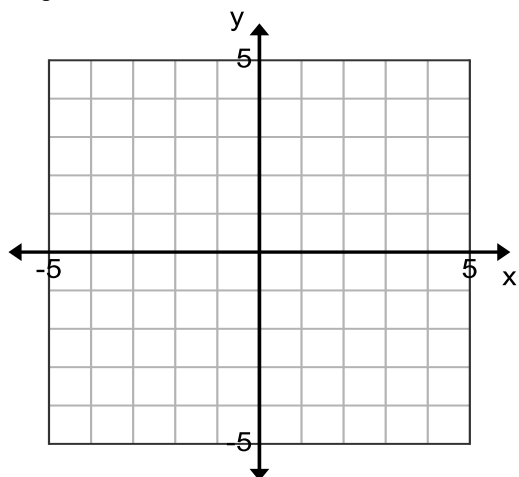
$y = \underline{\hspace{2cm}}$

$$f(x) = 3 - 2(x + 1)^3$$



13. Consider the function $f(x) = \begin{cases} 2x + 5 & \text{if } x < -2 \\ |x| - 1 & \text{if } -2 \leq x < 1 \\ (x - 2)^2 & \text{if } x \geq 1 \end{cases}$ (a) evaluate $f(4), f(-5), f(-2), f(1)$

b) Graph



14. Let $f(x) = \frac{1}{x-2}, g(x) = \frac{1}{x}$ and $h(x) = |2x + 3|$. Find each function and its domain.

(a) $(f \circ h)(2)$

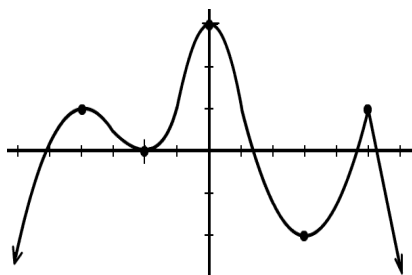
(b) $(f \circ g)(x)$

15. Find inverse function f^{-1} and state the domain and range for the inverse function

a) $f(x) = \frac{3-x}{2x+5}$

b) $f(x) = -\sqrt{9-x}$

16. The graph of a function g is shown.



State the intervals on which f is increasing and on which it is decreasing. Identify the **values** of any relative maximum or relative minimum.
(Relative min is 4 at $x=0$, it is not an order pair)

Increasing: _____

Decreasing: _____

Relative maximum: _____

Relative minimum: _____

17. Your college newspaper, The Collegiate Investigator, has fixed production costs of \$70 per edition, and marginal printing and distribution costs of 40¢/copy. The Collegiate Investigator sells for 50¢/copy.

a) Write down the associated cost, revenue, and profit functions.

b) What profit (or loss) results from the sale of 500 copies of The Collegiate Investigator?

c) How many copies should be sold in order to break even?